

多エージェントシステム

協調知能

MULTI-AGENT SYSTEM ARCHITECTURE

複数の自律エージェントが協調し、
知識を共有しながら複雑なタスクを
決めます。

LEGEND

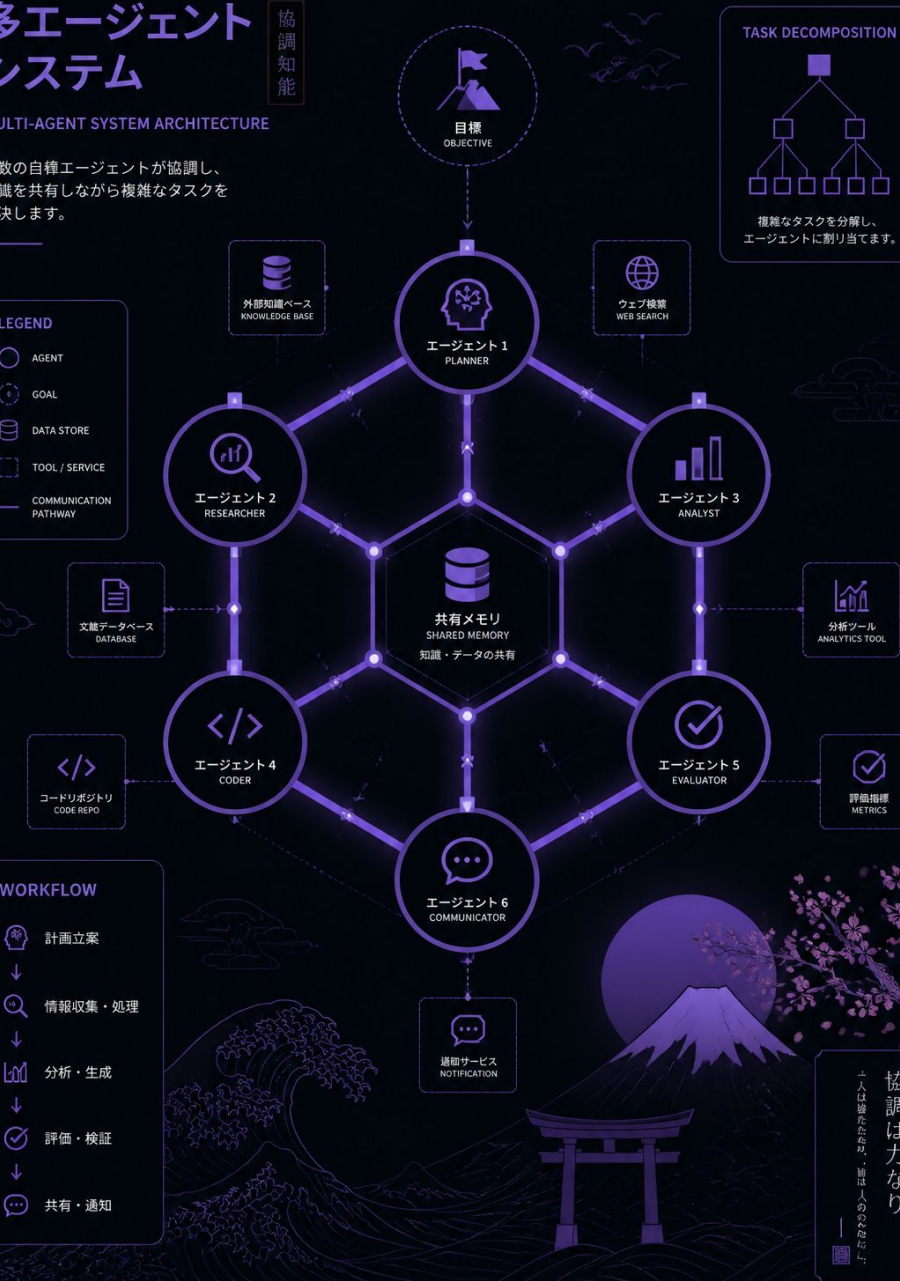
- AGENT
- GOAL
- DATA STORE
- TOOL / SERVICE
- COMMUNICATION PATHWAY

文蔵データベース
DATABASE

</>
コードリポジトリ
CODE REPO

WORKFLOW

- 計画立案
- ↓
- 情報収集・処理
- ↓
- 分析・生成
- ↓
- 評価・検証
- ↓
- 共有・通知



ENGINEERING MASTERCLASS 001

Career Intelligence Copilot FR-016: Multi-Agent Orchestration

How authority boundaries — not role names — justify when to introduce multiple AI specialists. An engineering deep-dive for experienced software and AI engineers.

The Engineering Problem

FR-015 already solved everyday preparation. FR-016 asked a fundamentally different question:

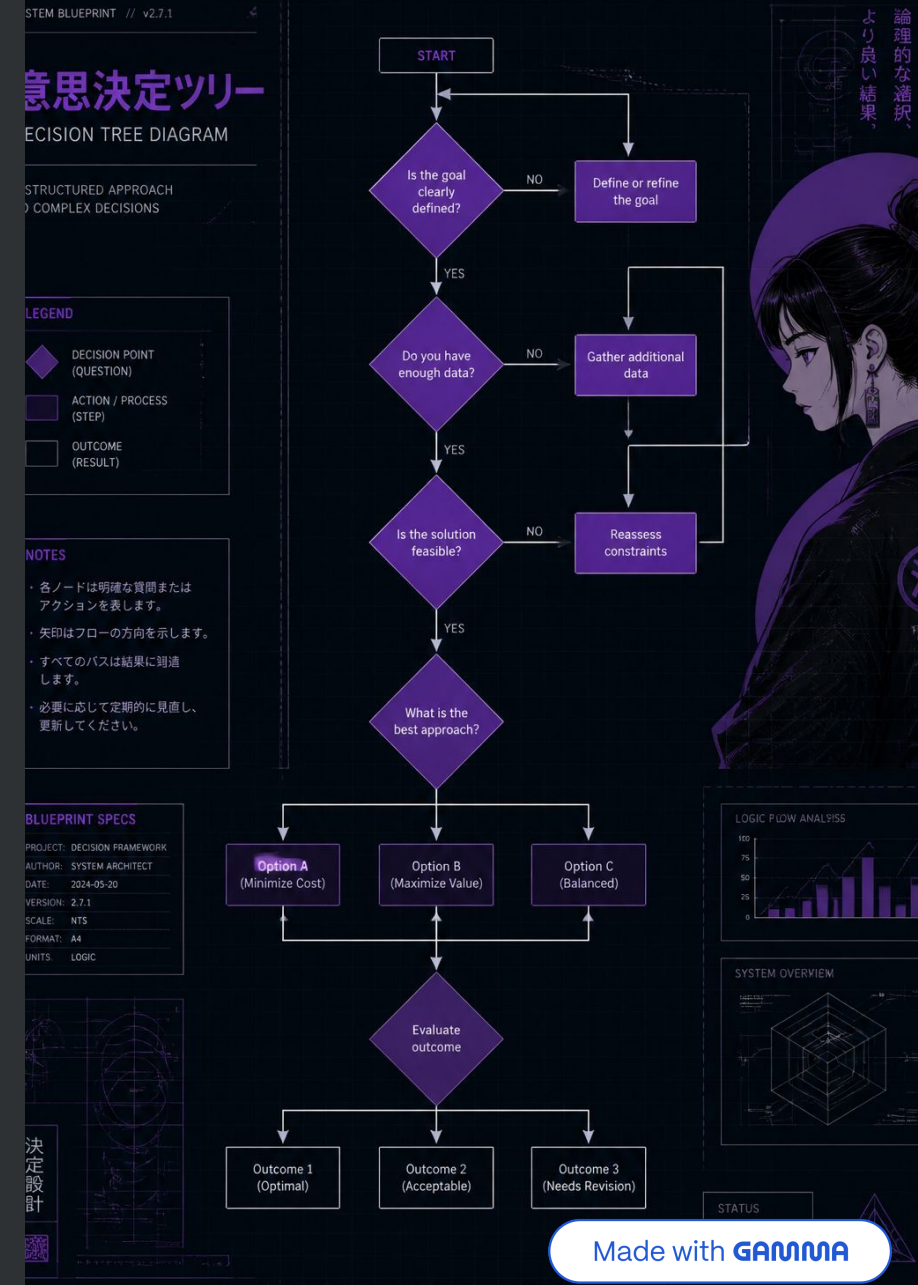
When does adding a second specialist create **real engineering value** — and when does it just add complexity?

FR-015 Solved

Everyday preparation with a single bounded agent. Simple, direct, effective.

FR-016 Asked

When is multi-agent orchestration *genuinely justified* by engineering constraints, not just appealing in theory?

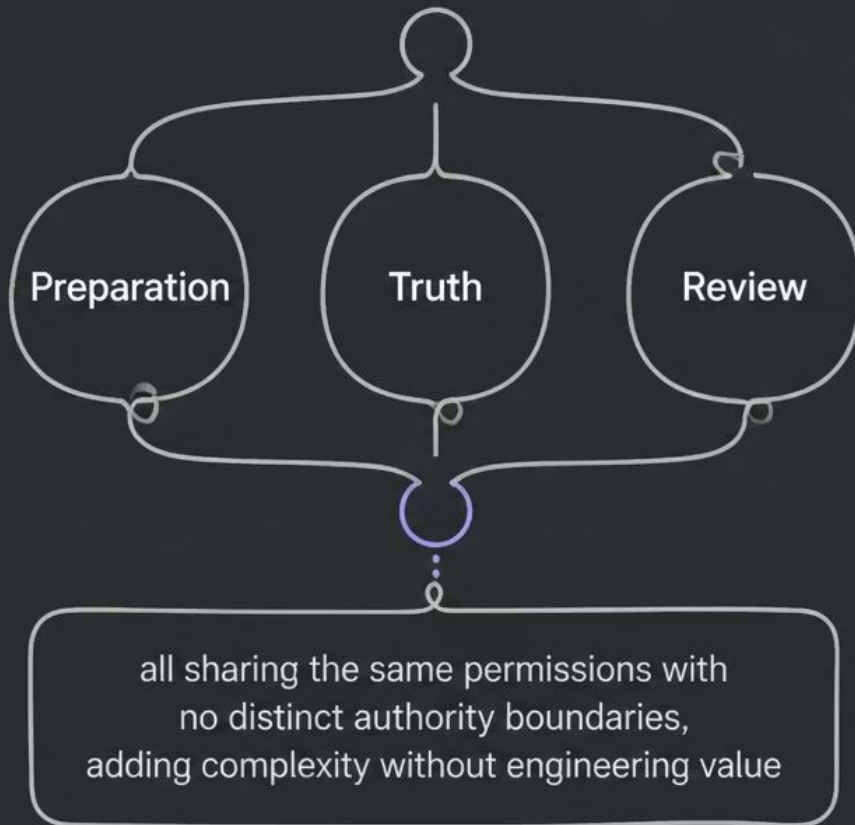


The Wrong Answer: "Multi-Agent Theatre"

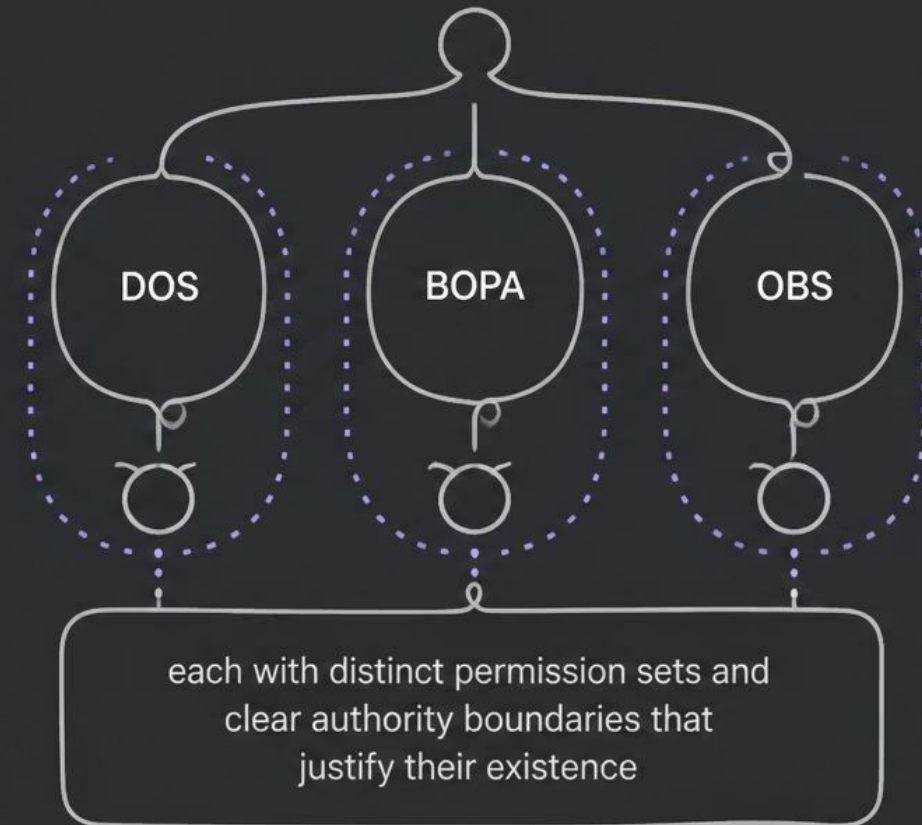
The tempting design: split by persona — Preparation Agent, Truth Agent, Review Agent. It feels sophisticated. It adds nothing real.

Comparison of two approaches:

Persona-Based Split (**rejected**)



Authority-Based Split (**accepted**)



⚠ **Key insight:** Role names are not authority boundaries. If every agent can do everything, you haven't added specialists — you've added indirection.

The Engineering Decision

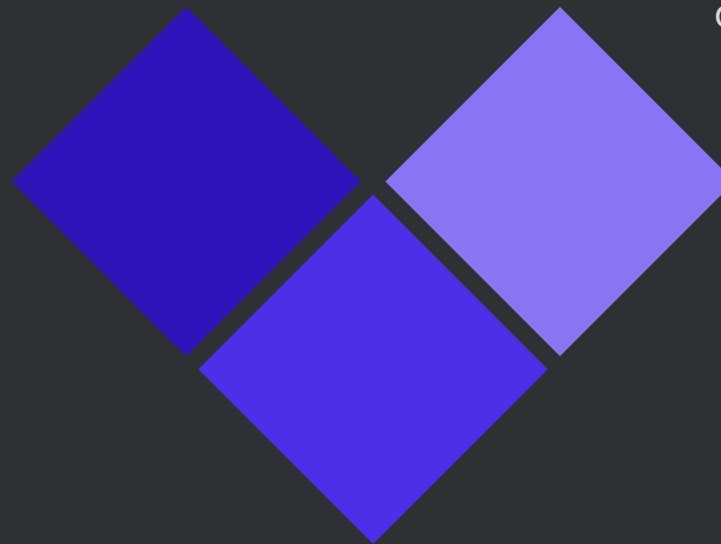
The accepted solution separates three concerns, each justified by a distinct **authority boundary** — not a job title.

Identify Authority Boundaries

Map distinct authority limits, not job roles.

Validate Unique Permissions

Ensure each specialist has others' exclusive access.

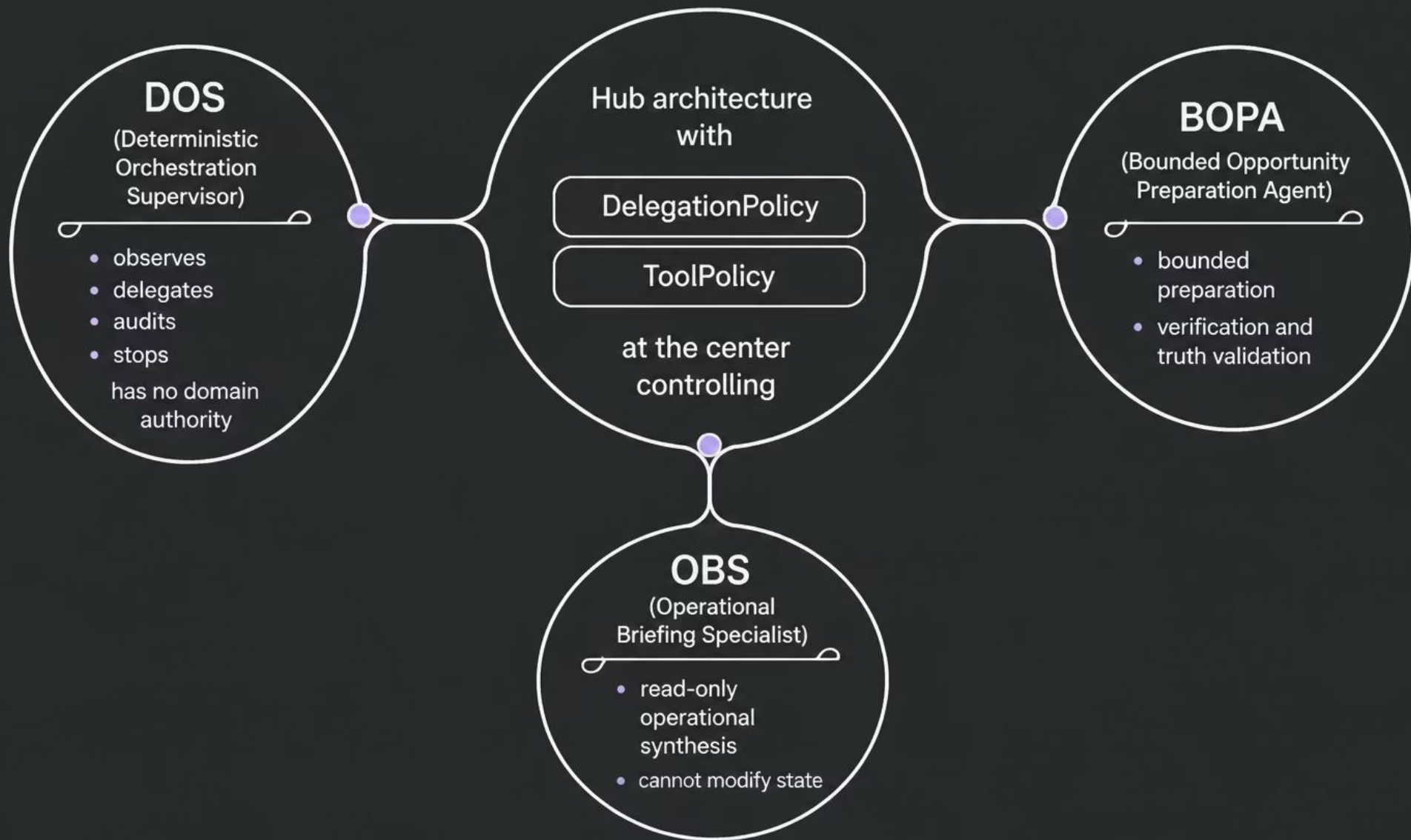


Assign One Specialist

DOS coordinates, BOPA prepares, OBS briefs.

Coordination, bounded preparation, and read-only synthesis are three genuinely different engineering concerns. Each requires different permissions. That difference is what justifies three specialists.

Final Architecture: DOS + BOPA + OBS



DOS

Coordinates without inheriting permissions. Observes, delegates, audits, stops.

BOPA

Bounded preparation, verification, and truth validation. The primary working agent.

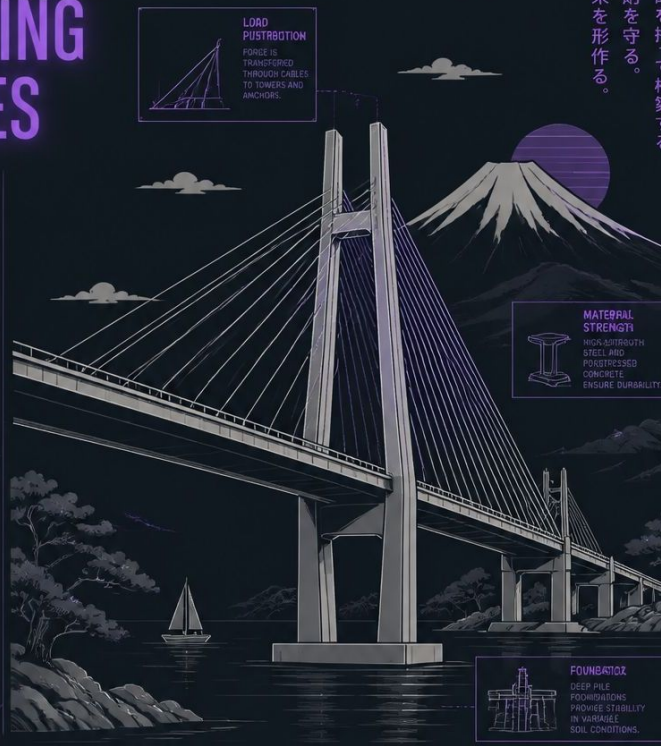
OBS

Read-only operational synthesis. Independent analysis that cannot modify assessed state.

ENGINEERING PRINCIPLES

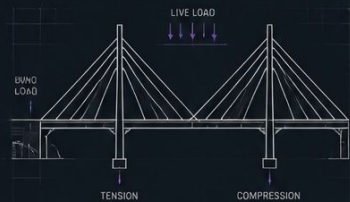
工学の原則

- SAFETY**
Design for protection.
Anticipate failure.
- FUNCTION**
Form follows purpose.
Every detail matters.
- STRENGTH**
Understand forces.
Build with integrity.
- EFFICIENCY**
Maximize impact.
Minimize waste.
- SUSTAINABILITY**
Respect the earth.
Build for the future.
- INNOVATION**
Question assumptions.
Create solutions.



原則を守る。
未来を形作る。

● STRUCTURAL ANALYSIS



● CABLE ANCHOR DETAIL



● DESIGN PROCESS



● MATERIAL COMPARISON

STEEL	STRENGTH	WEIGHT	DURABILITY
CONCRETE	STRENGTH	WEIGHT	DURABILITY
COMPOSITE	STRENGTH	WEIGHT	DURABILITY

改善
KAIZEN

"Continuous improvement
is the foundation of
lasting engineering."
— Build. Learn. Evolve.



Key Engineering Principles

Authority boundaries justify specialists

Distinct permissions — not distinct responsibilities — are the engineering justification for introducing another agent.

Supervisors coordinate, not execute

A supervisor that inherits domain permissions collapses the boundary it was meant to enforce.

Typed handoffs beat conversational delegation

Structured state transfers are safer and more auditable than natural-language routing between agents.

Deterministic orchestration when routing is knowable

If the routing decision can be expressed from typed state, make it deterministic — not learned.

Trade-offs & Validation

What Was Validated

- Safe deterministic orchestration
- Permission isolation between specialists
- Typed handoff safety
- Read-only OBS independence

What Was Intentionally Limited

- Commercial value was not the goal
- Direct BOPA remains the preferred daily workflow
- Orchestration is a validated foundation, not a product launch

❏ **Engineering outcome:** Success. **Product outcome:** Intentionally limited. Technical success does not automatically justify product adoption.

Why Employers Care About These Concepts

FR-016 demonstrates engineering judgment that goes beyond implementation skill. These are the signals interviewers are actually testing.



Authority Separation

Can you design systems where components have genuinely different permissions — not just different names?



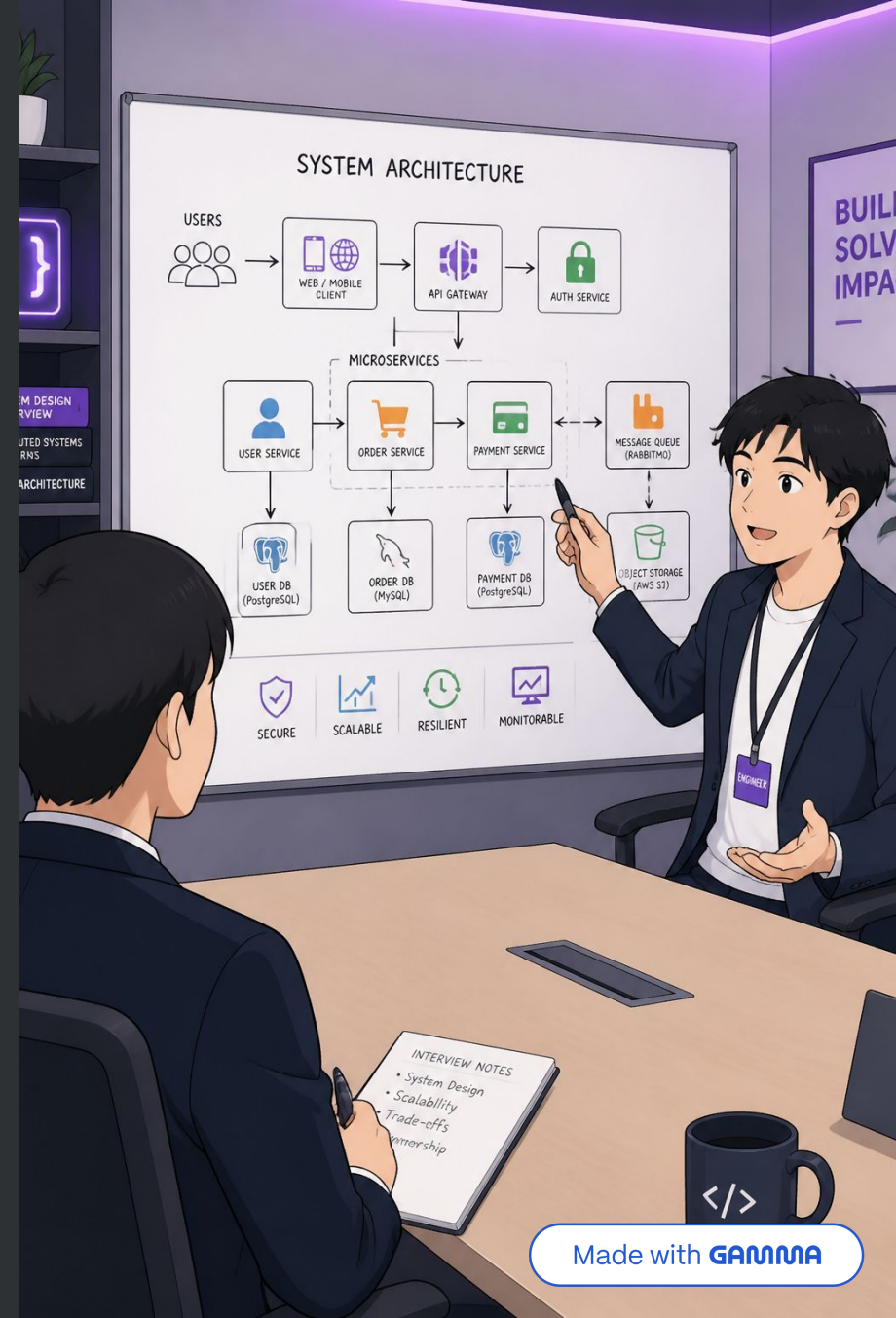
Choosing Determinism

Do you reach for ML when a typed-state decision is sufficient? Knowing when *not* to use AI is a senior skill.



Complexity Discipline

Can you justify every layer of indirection? Simpler architectures remain preferable until complexity earns its place.



Interview Preparation

These are the four questions FR-016 prepares you to answer confidently.

1 Why introduce multiple agents?

To separate **authority**, not merely responsibilities. Authority boundaries are the engineering justification.

2 Why is DOS deterministic?

Routing was fully expressible from typed state. Determinism improves safety and repeatability over learned routing.

3 Why is OBS read-only?

Independent analysis is stronger when it **cannot modify** the state being assessed. Write access corrupts objectivity.

4 Why wasn't orchestration made the default?

Direct BOPA remained simpler for everyday preparation. Added complexity must earn its place.

Three Things To Remember

① Authority Boundaries

Specialists are justified by **authority boundaries** — distinct permissions — not by role names or responsibilities.

② Coordinate, Don't Execute

Supervisors **coordinate** rather than execute. A supervisor that inherits domain permissions collapses the boundary it enforces.

③ Simpler Is Preferable

Simpler architectures remain preferable **until additional complexity earns its place**. Technical success $\neq$ product adoption.

- 📄 **FR-016 Milestone:** GO AS LEARNING PROOF ONLY — direct BOPA remains the preferred daily workflow while orchestration provides a validated architectural foundation.

